Context approach

6.4 Topic 4: Biodiversity and natural resources

Students will be assessed on their ability to:

1 Demonstrate knowledge and understanding of the practical and investigative skills identified in numbers 4 and 5 in the table of *How Science Works* on page 12 of this specification.

13 Explain the terms biodiversity and endemism and describe how biodiversity can be measured within a habitat using species richness and within a species using genetic diversity, e.g. variety of alleles in a gene pool.

14 Describe the concept of niche and discuss examples of adaptation of organisms to their environment (behavioural, physiological and anatomical).

15 Describe how natural selection can lead to adaptation and evolution.

2 Compare the ultrastructure of plant cells (cell wall, chloroplasts, amyloplasts, vacuole, tonoplast, plasmodesmata, pits and middle lamella) with that of animal cells.

3 Compare the structure and function of the polysaccharides starch and cellulose including the role of hydrogen bonds between β-glucose molecules in the formation of cellulose microfibrils.

5 Compare the structures, position in the stem and function of sclerenchyma fibres (support) and xylem vessels (support and transport of water and mineral ions).

4 Explain how the arrangement of cellulose microfibrils in plant cell walls and secondary thickening contribute to the physical properties of plant fibres, which can be exploited by humans.

7 Identify sclerenchyma fibres and xylem vessels as seen through a light microscope.

8 **Describe how to determine the tensile strength of plant fibres practically.**

9 Explain the importance of water and inorganic ions (nitrate, calcium ions and magnesium ions) to plants.

10 **Describe how to investigate plant mineral deficiencies practically.**

6 Describe how the uses of plant fibres and starch may contribute to sustainability, e.g. plant-based products to replace oil-based plastics.

12 Compare historic drug testing with contemporary drug testing protocols, e.g. William Withering's digitalis soup; double blind trials; placebo; three-phased testing.

11 **Describe how to investigate the antimicrobial properties of plants.**

16 Discuss the process and importance of critical evaluation of new data by the scientific community, which leads to new taxonomic groupings (i.e. three domains based on molecular phylogeny).

17 Discuss and evaluate the methods used by zoos and seedbanks in the conservation of endangered species and their genetic diversity (e.g. scientific research, captive breeding programmes, reintroduction programmes and education).

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